

Data Analysis & Visualisation Contribution

Part A: Introduction

Understanding the factors contributing to student depression has become increasingly important in recent years due to rising mental health concerns among university populations. Academic environments present a unique combination of stressors, including performance expectations, financial burdens, social pressures, and lifestyle changes. These factors can interact in complex ways, making it difficult to identify clear causes without structured analysis. This project aims to address this challenge by integrating data analysis and machine learning techniques to uncover patterns associated with student depression.

The primary objective of this work is to explore how different variables—such as academic pressure, financial stress, sleep habits, and lifestyle indicators—relate to depression levels among students. Through systematic data analysis and visualisation, the project seeks to provide meaningful insights that can support predictive modelling and inform potential interventions. By transforming raw data into interpretable visual formats, this contribution enhances understanding of underlying trends and relationships that may not be immediately apparent.

In addition, the use of visual analytics plays a critical role in bridging the gap between complex data and actionable knowledge. Stakeholders such as educators, policymakers, and mental health professionals can benefit from clear and accessible representations of data. Therefore, this report highlights not only the technical processes involved in data analysis but also the broader significance of these insights in addressing student mental health challenges.

Part B: Technical Review

Data analysis formed the foundation of this project, enabling the exploration and interpretation of relationships within the dataset. Several analytical techniques were employed to examine the structure, distribution, and interdependencies of variables.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis was the first step in understanding the dataset. This process involved summarising key characteristics of the data using statistical measures such as mean, median, standard deviation, and range. EDA also included identifying missing values, detecting outliers, and understanding the overall distribution of each variable.

Correlation Analysis

Correlation analysis was conducted to measure the strength and direction of relationships between variables. A correlation matrix was generated and visualised using a heatmap, allowing for quick identification of strong positive or negative relationships.

Data Visualisation Techniques

Visualisation played a key role in translating analytical findings into intuitive formats. Several types of visualisations were utilised:

- **Heatmaps** to represent correlation matrices
- **Histograms and density plots** to show data distributions
- **Box plots** to identify outliers and compare groups
- **Bar charts** to illustrate categorical relationships

These visual tools enabled a deeper understanding of the dataset and supported the identification of trends that informed machine learning model development.

Feature Selection Support:The insights gained from data analysis directly influenced feature selection. By identifying variables with strong relationships to depression, the model could focus on the most relevant predictors. This not only improved model efficiency but also enhanced interpretability.

Part C: Design & Implementation

Individual Contribution: Data Analysis and Visualisation: My primary responsibility in this project was to conduct data analysis and develop visualisations that would support both understanding and modelling efforts. This involved working closely with the dataset to uncover meaningful patterns and presenting findings in a clear and structured manner.

Data Preparation

Before analysis, the dataset required preprocessing to ensure accuracy and consistency. This included:

- Handling missing values through imputation or removal
- Normalising numerical features where necessary
- Encoding categorical variables into numerical formats
- Removing duplicate or inconsistent entries

Proper data preparation was essential to ensure that subsequent analysis and visualisations were reliable.

Visualisation Development: A range of visualisation techniques were implemented to explore different aspects of the data:

Heatmaps: Heatmaps were used to visualise the correlation matrix, providing a comprehensive overview of relationships between all variables. This allowed for quick identification of key influencing factors.

Distribution Plots:- Distribution plots were created to analyse how individual variables were spread. These plots revealed important insights, such as the variability in sleep patterns and differences in stress levels among students.

Comparative Visualisations:-Box plots and grouped bar charts were used to compare depression levels across different categories, such as varying levels of academic pressure or financial stress. These comparisons helped highlight disparities and identify high-risk groups.

Key Findings

The analysis revealed several significant relationships between variables and depression levels:

- **Academic Pressure:** A strong positive relationship was observed, indicating that higher academic demands are associated with increased depression levels.
- **Financial Stress:** Students experiencing financial difficulties showed higher levels of depression, suggesting that economic factors play a critical role in mental health.
- **Sleep Patterns:** Poor sleep quality and irregular sleep schedules were linked to higher depression scores, highlighting the importance of healthy sleep habits.

These findings were consistent across multiple visualisations, reinforcing their reliability and significance.

Integration with Machine Learning: The insights gained from data analysis were used to inform machine learning models. Features identified as highly correlated with depression were prioritised during model training. This contributed to improved model performance and interpretability.

Part D: Technical & Critical Evaluation

Technical Evaluation

The data analysis and visualisation process significantly enhanced the overall effectiveness of the project. By providing clear insights into the dataset, visualisations supported both feature selection and model validation.

One of the key strengths of this approach was its ability to improve interpretability. Machine learning models can often be perceived as "black boxes," but visual analysis helps explain why certain predictions are made. For example, the strong correlation between academic pressure and depression provided a clear rationale for model outcomes.

Additionally, the alignment between correlation patterns and machine learning results increased confidence in the model's reliability. When both analytical and predictive approaches point to similar conclusions, the findings are more robust.

However, there were also limitations. Visualisations can sometimes oversimplify complex relationships, and improper scaling or design choices may lead to misinterpretation. Therefore, careful attention was required to ensure accuracy and clarity.

Critical and Social Evaluation

While the project produced valuable insights, it is important to consider its limitations and broader implications.

Dataset Limitations

The dataset used in this project may not fully represent the diversity of student populations. Factors such as cultural background, geographic location, and socioeconomic status were not comprehensively captured. As a result, the findings may not be generalisable to all contexts.

Missing Contextual Factors

Depression is influenced by a wide range of factors beyond those included in the dataset. For example, personal relationships, mental health history, and environmental conditions can all play significant roles. The absence of these variables limits the scope of the analysis.

Ethical Considerations

Working with mental health data requires careful ethical consideration. Privacy and confidentiality must be maintained at all times. Additionally, predictive models should be used responsibly to avoid stigmatization or misuse.

Risk of Misinterpretation

Visualisations can be powerful but may also lead to incorrect conclusions if not interpreted properly. For instance, a strong correlation might be mistaken for a causal relationship. It is essential to communicate findings clearly and highlight any uncertainties.

Part E: Project Proposal & Management

Reflection: This project demonstrated the value of data visualisation in understanding complex datasets. Visual tools made it easier to identify patterns and communicate findings effectively. However, the process also highlighted the importance of critical thinking when interpreting results.

Several challenges were encountered during the project:

- Handling incomplete or inconsistent data
- Selecting appropriate visualisation techniques
- Balancing detail with clarity in visual representations
- Ensuring that findings were accurately interpreted

Overcoming these challenges required iterative analysis, experimentation, and validation.

Future Work

To further enhance this project, several improvements are proposed:

- **Interactive Dashboards:** Developing interactive visualisations would allow users to explore the data dynamically and gain deeper insights.
- **Real-Time Data Analysis:** Incorporating real-time data could provide up-to-date insights and improve the relevance of predictions.
- **Advanced Visualisation Techniques:** Techniques such as clustering visualisations and dimensionality reduction could reveal more complex patterns.

These enhancements would strengthen both the analytical and practical value of the project.

Part F: Conclusion

Data analysis and visualisation played a crucial role in this project by transforming raw data into meaningful insights. Through techniques such as correlation analysis and graphical representation, key factors influencing student depression were identified and understood.

The findings highlighted the significant impact of academic pressure, financial stress, and sleep patterns on mental health. These insights not only supported machine learning model development but also provided a foundation for potential interventions.

Despite its limitations, the project demonstrates the power of combining data analysis with visualisation to address complex real-world problems. By continuing to refine these methods and expanding the scope of analysis, future work can further contribute to improving student well-being and mental health support systems.

Overall, this contribution underscores the importance of data-driven approaches in understanding and addressing student depression, offering valuable insights for both academic research and practical applications.

SHU AI Transparency Scale (AITS)

Descriptor:- AI for Shaping

Transparency Statement: AI has been used to shape the initial and or final parts of the activity.

AI Contributions:- AI is used for shaping parts of the activity. This includes initial outlining, concept development, prompting thinking, and improving the structure quality of the final output

Human Contribution: Most of the activity is human-developed and generated. AI ideas and suggestions are refined and reviewed. AI outputs are used for discrete and specific goal outcomes.